

Electrocoagulation Arduino Setup

Of course! Let's **break down** your final code clearly so you fully understand it:

Code Explanation:

cpp

```
#include <AFMotor.h>
```

- Loads the **AFMotor** library to control motors with your L293D motor shield.

cpp

```
AF_DCMotor motor(1, MOTOR12_64KHZ);  
AF_DCMotor pump(2, MOTOR12_2KHZ);
```

- Defines two motors:
 - **motor** (for the **electrodes**) connected to **Motor 1** (high frequency 64kHz PWM).
 - **pump** (for the **water pump**) connected to **Motor 2** (2kHz PWM).

cpp

```
unsigned long startTime;  
bool ecDone = false;
```

- **startTime** : Will save the moment when the electrocoagulation (EC) started.
- **ecDone** : A flag (true/false) that tells the Arduino if the EC phase has finished.

cpp

```
void setup() {  
  Serial.begin(9600);  
  Serial.println("Starting Electrocoagulation...");  
  
  motor.setSpeed(255);  
  pump.setSpeed(255);  
  
  startTime = millis();  
}
```

- `Serial.begin(9600)` : Starts the Serial Monitor (optional for checking).
- Set motor **speed** to **maximum** (255 out of 255).
- Save the current time (`millis()`) to know when EC started.

cpp

```
void loop() {  
  unsigned long currentTime = millis();
```

- In the `loop()` , Arduino keeps checking the **current time**.

Inside `loop()` :

1. Electrocoagulation Phase (20 min):

cpp

```
if (!ecDone) {  
  if (currentTime - startTime < 1200000) {
```

- If EC has **not finished**, and time passed is **less than 20 minutes (1200000 ms)**:
 - Alternate motor movement:

- Run **FORWARD** for 20 seconds.
- Run **BACKWARD** for 20 seconds.

cpp

```
motor.run(FORWARD);  
Serial.println("Electrodes FORWARD");  
delay(20000);  
  
motor.run(BACKWARD);  
Serial.println("Electrodes BACKWARD");  
delay(20000);
```

2. After 20 min (EC Finished):

cpp

```
else {  
    motor.run(RELEASE);  
    ecDone = true;  
    Serial.println("Electrocoagulation Done. Starting Pump...");  
    pump.run(FORWARD);  
}
```

- After 20 min:
 - **Stop** the electrodes (RELEASE).
 - **Mark** EC as done.
 - **Start** the pump running **continuously**.

3. After EC Done (Pump Phase):

cpp

```
else {  
    pump.run(FORWARD);
```

```
}
```

- Keep the **pump running forever** after EC finishes.
- No delay, no stopping.

⚡ Short Summary:

Phase	Action
Electrocoagulation	Electrodes alternate 20 sec forward / 20 sec backward for 20 min
After EC finished	Electrodes stop, pump turns on and stays on forever

📌 Important:

- `millis()` function gives the time in milliseconds since the Arduino started. (No need for complicated timers.)
- Using **flags** like `ecDone` is a professional way to **separate phases** in Arduino projects.
- **No sensors** are used (only time-based control).
- Very clean, simple, and ready for real-world testing.

Would you also like me to help you write a small paragraph explaining this code **in a scientific way** for your report or poster? 🚀 (e.g., "Arduino programming was used to control the timing of the electrocoagulation process...")